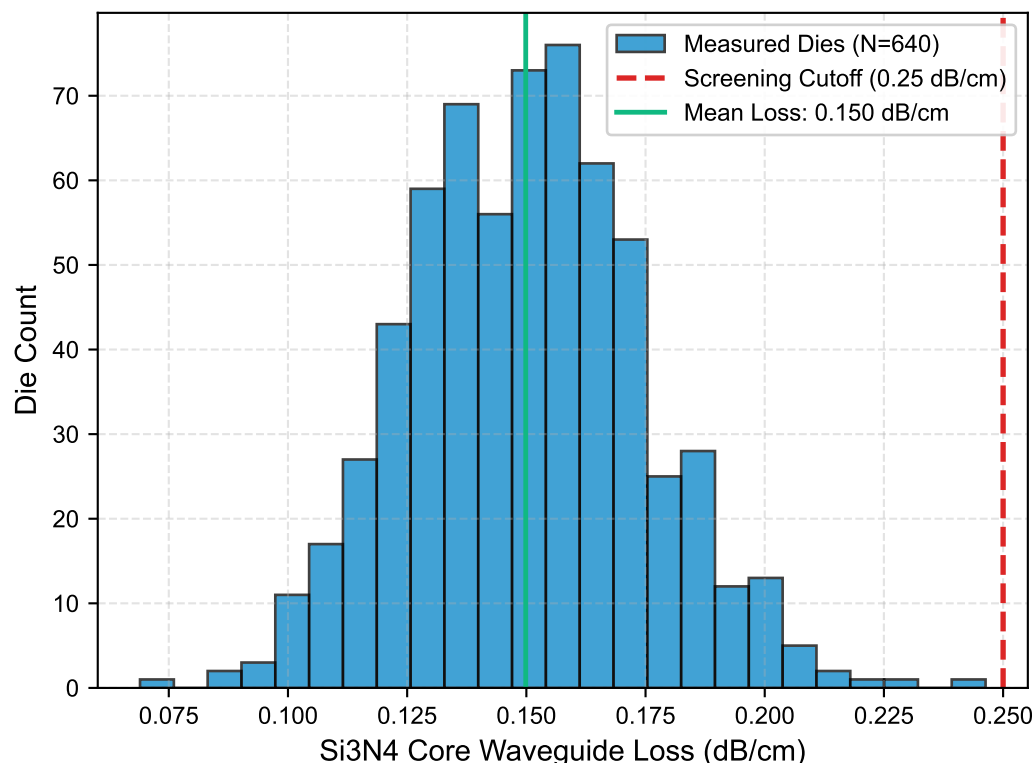


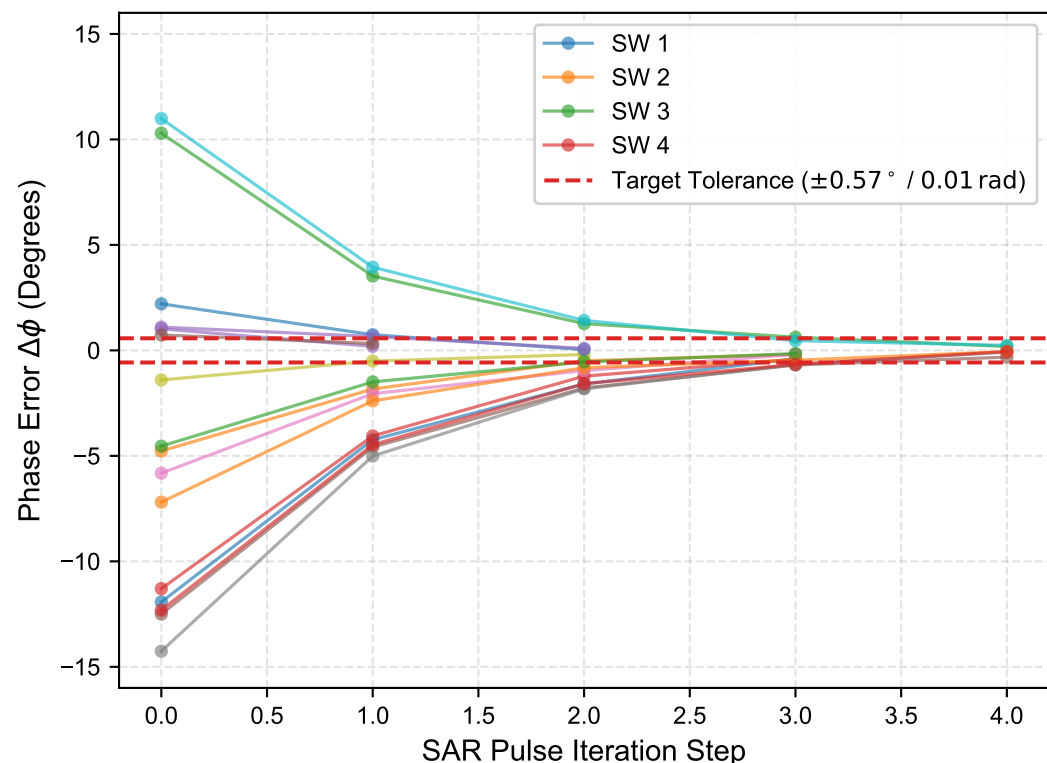
PROJECT JANUS MINI-16: UNIFIED ELECTRO-PHOTONIC DFT & BIST SIGN-OFF MATRIX

Photonic Loopback Probing (P-BIST), PCM State Trimming, APD DAC, IEEE 1500 Scan & RRNS Self-Checking

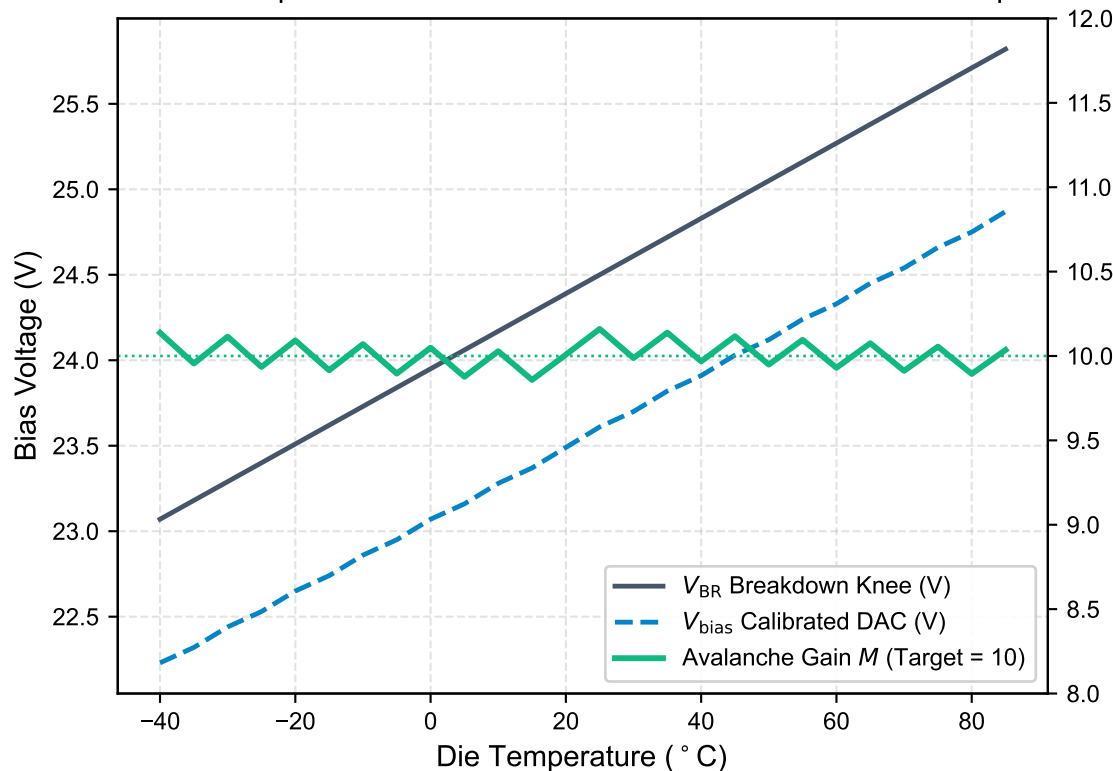
(a) Wafer-Level Optical Loopback Screening (P-BIST)
Yield: 100.0% (640/640 Passed) | Grating Coupler: 2.12 dB



(b) Sb2S3 PCM Closed-Loop Phase Calibration
All 16 Switches Converged | Max Error: 0.0073 rad | Settle Time: 1.40 μ s



(c) Dynamic APD Bias Tracking DAC (-40 C to +85 C)
Fixed Multiplication Gain $M = 10.0 \pm 0.16$ Across Thermal Envelope



(d) IEEE 1500 Scan ATPG & RRNS Online BIST
4,096 Scan Cells | 40,960 Nodes | Coverage: 99.85%

